

7

Mobile Web Surveys

A Total Survey Error Perspective

Mick P. Couper,^{1,2} Christopher Antoun,^{*,3} and Aigul Mavletova⁴

¹ Survey Research Center, University of Michigan, Ann Arbor, MI, USA

² Joint Program in Survey Methodology, University of Maryland, College Park, MD, USA

³ Center for Survey Measurement, U.S. Census Bureau, Suitland, MD, USA

⁴ Department of Sociology, National Research University Higher School of Economics, Moscow, Russia

7.1 Introduction

While mobile phones or cell phones have been a challenge for telephone survey researchers for some time, the Internet or web capabilities of mobile phones have begun to receive attention in the last few years. There are a number of ways that Internet-enabled smartphones can affect survey data collection, and the implications of these for various sources of errors are only now being fully explored. There are three broad approaches to the opportunities and challenges posed by mobile web.

The first focuses on the many opportunities that such devices can bring to extend and enhance measurement (see AAPOR, 2014). These include the promise of “anytime, anywhere” or “in-the-moment” measurement, in which respondents can be prompted to provide survey responses at set times or based on location. The enhanced capabilities of smartphones allow respondents to capture and transmit pictures. Barcode and quick response (QR) code scanners also allow respondents to provide detailed information on actions they may have taken. Similarly, global positioning systems (GPS) give researchers the opportunity to record respondents’ locations and movements. Accelerometers and similar capabilities facilitate the measurement of physical activity. All of these permit the researcher to go beyond the traditional measurement tools of questions and answers. They also present a number of methodological challenges, including those relating to respondent consent, privacy, and measurement bias. Selection bias remains the biggest concern as these technologies are extended beyond small-scale feasibility studies among volunteers, to large-scale population-based studies.

A second broad approach to the use of mobile web relates to the sampling opportunities it may provide. For example, given that such devices are both phones and Internet-enabled

*Part of this work was done while the author was at the University of Michigan. Any views expressed are those of the author and not necessarily those of the U.S. Census Bureau.

devices, it may be possible to use RDD-based sampling methods to overcome the challenge of a lack of frame or the unstructured format of e-mail addresses to serve as a sampling frame. Of course, there are a number of practical challenges to this approach, relating to both coverage (those who do not have a smartphone) and nonresponse (those unwilling or unable to complete a web survey), not to mention the legal restrictions related to the sending of unsolicited text messages. Further, to the extent that mobile users increasingly rely on alternatives to telephone numbers to communicate through social media and texting, using this approach as a sampling frame may become more limited. In similar fashion, mobile web is viewed as potentially overcoming the coverage problems associated with Internet access. To the extent that mobile web users are different from traditional Internet users, including mobile-only web users in samples may reduce coverage bias. This approach views the mobile web problem in similar vein to the cell phone challenge faced by telephone survey researchers.

A third approach to mobile web focuses on respondents who complete web surveys using mobile devices, whether of their own volition or by design. A number of early observational studies reported on the challenges—such as higher breakoff rates and longer completion times—when completing web surveys on mobile devices. More recently, experimental designs are exploring the nonresponse and measurement error implications of mobile web use.

Our primary focus is on the coverage and nonresponse implications of including or excluding mobile web users, and the measurement error implications of web survey completion on mobile devices. Given the rapid rise of this new “mode” of data collection and the variety of devices that can be used to answer online surveys, we need to understand the implications of these design options from a total survey error perspective. In the remainder of this chapter, we review the current state of knowledge with regard to mobile web. We first describe more fully what we mean by “mobile web.”

In the literature, a distinction is made between PCs (desktop and laptop computers) on the one hand and mobile devices (including tablets, feature phones,¹ and smartphones) on the other. Some researchers combine tablets with PCs, while others group them with smartphones.² Given the continued proliferation of device types, we believe it is important to understand the characteristics or features of the devices that have implications for sources of errors. A key dimension is the size of the device, and particularly the size of the display screen. Given the proliferation of different devices, this should be thought of as a continuum rather than a dichotomy. While tablets and smartphones differ in size, they have other features in common.

A second set of dimensions relates to the technology features of the devices. These may include the make and model, operating system, processing power, connectivity (both speed and reliability; cellular vs. Wi-Fi connections), and data entry interface (touchscreen, onscreen keyboard, etc.). Some of these affect the designer’s ability to deliver a consistent survey experience across systems. Others, like transmission time and reliability, may affect survey completion and survey length. The mode of input may affect responses to questions requiring open-ended responses. All of these technology features have potential implications for our ability to survey people successfully using these devices, and the quality of the resulting data.

A third dimension of mobile web relates to user characteristics and behavior. These include such factors as familiarity or comfort with the device, how respondents use the device (e.g., frequency of use, duration of browsing sessions), how they pay for device use, the availability of

1 A cell phone that contains a set of functions beyond voice calling and text messaging, but not as extensive as a smartphone. Feature phones may offer web browsing and e-mail, but generally cannot download or run apps.

2 We use the term “mobile device” to refer more broadly to both types of devices, and “smartphone” to refer specifically to cell phones with enhanced capabilities.

alternative devices, and the like. Again, these may affect respondents' willingness and ability to complete a mobile web survey.

A fourth dimension is context of use. Mobile devices are used in a variety of different settings. Understanding the contexts in which people use their mobile devices for various activities is important for explaining possible differences in survey errors. Not only are physical mobility and location important, but so too are related factors such as the degree and type of distractions, the presence of others, or multitasking behavior—whether on the same device, on a different device, or even in a different medium.

Researchers are dealing with mobile devices in a number of different ways (see Chapter 12), ranging from ignoring the problem or discouraging mobile use, to more active approaches to accommodating mobile users, whether through mobile optimization or through the use of apps. Our focus here is primarily on browser-based surveys (i.e., not apps or app-like approaches; see Buskirk and Andrus, 2012), whether or not they are optimized for smartphone use. This is currently the most common approach to dealing with mobile web users.

In the next three sections (Sections 7.2 through 7.4), we focus on the coverage, nonresponse, and measurement error implications of mobile web surveys. It is oftentimes difficult to distinguish between coverage and nonresponse error: invitations to a mobile web survey may go unheeded, either because sample persons do not have access to a mobile device with Internet access (non-coverage) or because they are unwilling or unable to use their devices to respond to the survey (non-response). In either case, selection bias may be the result. We discuss coverage and nonresponse error in turn below, but note that the distinction may be somewhat artificial (see Chapter 5).

7.2 Coverage

The different ways in which researchers deal with mobile devices have implications for who is observed and who is not observed, that is, coverage errors. This is a function of both the proportion of the target population that has access to the required device and the differences between those with access and those without access. Researchers have begun exploring whether using mobile devices (or not using them) in web surveys designed to make inference to a broad population leads to biased survey estimates. There are two different perspectives on this matter of coverage in mobile surveys.

The first focuses on coverage issues in mobile-only web surveys that require respondents to install a survey app on their phones or use some other mobile-only features (e.g., GPS, Bluetooth-enabled sensors). Those without smartphones could never participate in such surveys unless they are provided phones, which may be cost prohibitive. While the literature on web survey coverage focused on the distinction between those with Internet access and those without (i.e., the digital divide), here the new distinction is the so-called device divide which distinguishes those who own (or have access to) a smartphone from those who do not (see Pearce and Rice, 2013). Surveys that ask about technology use in the U.S. and Europe report smartphone coverage rates ranging from 18 to 64%, depending on the year and country (Fuchs and Busse, 2009; Smith, 2015; Statistics Netherlands, 2013). They also include the issue of how those on one side of the device divide (with mobile access) differ from those on the other side (those without access). Using U.S. national telephone surveys, Pew researchers have found that those with smartphone access tend to be younger, better educated, and more likely to be Black or Hispanic compared to those without access (Smith, 2015). Using a face-to-face survey conducted in 33 European countries (Eurobarometer), Fuchs and Busse (2009) also report uneven access across demographic groups. They find that those with mobile Internet access tend to be younger, more likely to

be male, more likely to be single, more likely to be living outside of a rural area, and older when they obtained their last educational degree. In the same study, they estimate coverage bias for five socio-demographic variables across 27 of these countries. The largest relative bias was for age (on average 69%) followed by age at highest degree (28%) and type of community (14%). Using Fuchs and Busse's approach (2009), Metzler and Fuchs (2014) use Eurobarometer data from 2013 to update these results in light of increasing mobile penetration rates. As expected, they find that mobile web bias estimates did decline with increasing mobile web penetration, but the gains were modest and not consistent across all demographic estimates. This suggests that limiting web surveys to mobile web users only is still risky for surveys targeted at general populations.

These results, however, must be considered in light of the current trends in mobile Internet coverage. It is estimated that mobile Internet access in the U.S. has more than doubled between 2009 and 2013, jumping from 21 to 55% (Duggan and Smith, 2013). Access in the EU also recorded notable growth between 2005 and 2013, increasing from 27 to 43% (Eurobarometer data retrieved from Metzler and Fuchs (2014)). If current trends continue then mobile coverage may become less of a concern over time, assuming that the differences between the covered and noncovered remain steady. However, the coverage problem is unlikely to completely disappear.

The other perspective focuses on whether including mobile users improves coverage of web surveys. Offering a mobile option may be effective in capturing hard-to-reach respondents who prefer using their smartphones. Toepoel and Lugtig (2014) compared the sample compositions obtained from a passive web questionnaire that did not accommodate mobile users and a responsive questionnaire that did accommodate such users. Offering the mobile option did help in reaching younger respondents (average age of 44) compared to the passive approach (average age of 48), though the gains were modest. This likely reflects the relatively small but increasing demand for a mobile option in web surveys from mobile-mostly and mobile-only users (who do not have landline Internet access). According to U.S. national telephone surveys conducted by Pew, 15% of smartphone owners have a "limited number" of other options for going online and 10% have no landline Internet access at home (Smith, 2015).

Respondents who prefer to use their phones, whether by choice or by necessity, may be different from those who prefer other devices. In an analysis of Pew data from a U.S. telephone survey, Antoun and Couper (2013) report that three substantive measures (trust in others, Internet use, and PC access) were significantly correlated with a preference for mobile web, even after controlling for demographic differences. In another multivariate analysis using Pew data, Antoun (2015a) reports that tablet use is associated with both smartphone web access and a preference for using a smartphone to go online. Similar results are found when looking at the actual use of mobile devices. In a web survey of smartphone owners using a U.S. probability-based online panel, Wells et al. (2013a) report that those who chose to respond using their phone were younger, more likely to be female, and more likely to live in larger households compared to PC respondents. In a web survey using a U.S. market research panel, Peterson (2012) also found that mobile respondents were younger and more likely to be female than PC respondents. In an analysis of device usage in the Dutch LISS panel, de Bruijne and Wijnant (2014a) similarly report that mobile and tablet respondents were younger, more likely to be female, and more likely to identify themselves as early adopters of new technology and as more progressive than PC respondents. This suggests that to minimize coverage error there is much to gain by adding mobile users, or conversely, much to lose by excluding them.

How people use their smartphones also has implications for coverage. A distinction should be made between possession of a device and ability to reach a respondent using that device. Owning an Internet-enabled smartphone is necessary to reach a respondent but not sufficient. Sample persons also have to use (and know how to use) mobile browsers.

Table 7.1 Response rates for PC and mobile web.

	Response rate (%)	
	PC web	Mobile web
De Bruijne and Wijnant (2013) ^{a)}	61	47
Mavletova (2013)	82	40
Mavletova and Couper (2013)	74	31
Wells et al. (2013a) ^{a)}	61	58
Buskirk and Andrus (2014)	64	23
Antoun (2015b) ^{a)}	85	74

All mobile questionnaires were optimized for small screens.

a) Probability panels; others are opt-in panels.

To be covered, sample members must also receive the invitation that is sent to them. Coverage is therefore linked not only with the mode of survey completion but also with the mode of invitation (e.g., text message (or SMS), mail, e-mail), and the quality of the information on the sampling frame. While absorption rates³ tend to be higher for e-mail than SMS (Mavletova and Couper, 2013), they are still less than 100% because e-mail invitations can end up in a spam filter or bounce back as undeliverable (see Callegaro and DiSogra, 2008).

A final consideration is household- versus individual-level coverage. Mobile phones are typically personal devices that often move with their users. According to Carley-Baxter et al. (2010), only about 15% of mobile-only users in the U.S. shared their device with other family members. Limited phone sharing can make it difficult to reach other people in the household via a smartphone compared to a PC, which is more likely to be shared. There are exceptions to this in some settings where mobile phones are often shared among friends and family members (see e.g., Steenson and Donner, 2009). In any case, this suggests the need to conceptualize access to mobile Internet at the individual level rather at the household level.

7.3 Nonresponse

Even if sample persons are capable of responding via mobile devices, it does not mean that they will necessarily choose to do so. In this section, we focus on two types of nonresponse of importance to mobile web surveys: unit nonresponse and breakoffs.

7.3.1 Unit Nonresponse

While accommodating mobile web users may improve coverage of mobile-mostly or mobile-only groups, this only reduces selection bias to the extent that people respond to surveys using mobile devices. However, several experimental studies report higher unit nonresponse for mobile web than PC web in both volunteer panels and probability panels (Table 7.1). The differences are sometimes quite large. One reason may be that some participants who are less familiar with mobile technology are instructed to use a smartphone even though they would not

³ The absorption rate is an indicator of the proportion of e-mail addresses (or text messages) that are successfully delivered (see Callegaro and DiSogra, 2008).

themselves use such a device when given a choice. However, these differences are observed even after extensive prescreening to identify those who report being willing and able to complete a survey on their smartphone.

Another possible reason for the higher nonresponse in mobile web than PC web relates to technology features of smartphones. The connection type (cell network vs. Wi-Fi), speed (fast vs. slow), reliability (intermittent vs. consistent), and device type (smartphone vs. feature phone) may have implications for how willing people are to use their phones to take surveys. De Bruijne and Wijnant (2014a) found that those who did not respond to a mobile web survey tended to have less advanced phones with navigation buttons rather than more advanced ones with full touchscreens. A further reason for differential nonresponse relates to smartphone user characteristics and their habits. This includes the degree of motivation and trust in completing mobile web surveys and whether such surveys fit naturally with the way that smartphones are typically utilized.

Bosnjak et al. (2010) report that perceived trustworthiness and perceived enjoyment in mobile web surveys are related to the intention to participate in them. De Bruijne and Wijnant (2014a) found that those who reported not reading e-mails on their smartphones were significantly more likely to use other devices or not respond at all to a mobile web survey. Mavletova and Couper (2014, 2016a) and Antoun (2014) all found that nonrespondents to their mobile web surveys tended to use their phones less frequently than respondents. Antoun also found that those who typically use their phones to go online for brief sessions (<2 minutes) responded at lower rates than those who browse for longer periods.

The use of multiple devices is also a potential deterrent to participation on a smartphone. Some multidevice users prefer to use their tablet or PC for longer tasks. This reflects the view that smartphones are “go-to devices” preferred for short activities, while PCs are “workhorse” devices preferred for longer tasks (see Robinson, 2014). Or, some respondents may be willing to complete surveys on their smartphone, but only in the evening when mobile Internet usage peaks rather than during the workday when they may rely more on a PC (Lipsman and Aquino, 2013). If smartphones are used for briefer browsing sessions than PCs (Cui and Roto, 2008), the stated survey length in the introduction may have stronger effects for mobile web users. Kelly et al. (2013) reported that fewer smartphone users in a volunteer web panel were willing to spend more than 15 minutes completing mobile web surveys (35%) compared to PC web surveys (81%).

Given that mobile web surveys are associated with higher levels of unit nonresponse, what can be done to increase the proportion of mobile web respondents? Similarly, if we are to exploit the advanced features of smartphones, how can we encourage more people to use mobile devices to complete surveys? Several recent studies show the growth in use of smartphones for web survey completion. For example, Sarraf et al. (2015) show smartphone use in the U.S. study called National Survey of Student Engagement (NSSE) growing from 4% in 2011 to 8% in 2012, 12% in 2013, 18% in 2014, and 27% in 2015. Saunders and Kessler (2015) report that mobile phone use in MaritzCX web surveys has also grown by an average of 10 percentage points per year from 2012 to 2015, increasing from 9 to 38% (see also Chapter 10). This suggests that researchers cannot ignore the rapid growth in survey participation via smartphones. That said, the proportion of people who choose to use smartphones to complete web surveys is still smaller than the proportion of people who have access to such devices (i.e., potential users), especially in probability-based panels. Poggio et al. (2015) show that smartphone use across eight waves of the GESIS Online Panel pilot ranged from only 1.8 to 2.5%. De Bruijne and Wijnant (2014a) report that the use of a smartphone to complete surveys in the CentERpanel in September 2013 was 1.6%. This low rate of smartphone use is likely related to some combination of under-coverage (see Section 7.2), unwillingness to use smartphones to complete web surveys, and non-response among those who intend to use their smartphones.

Efforts to encourage smartphone use among respondents have been moderately successful. In an experiment where college students were prompted to use a smartphone to complete a web survey, Millar and Dillman (2012) report that about 15% of smartphone owners did so. In Toe-poel and Lugtig's (2014) more recent study that used a mobile-optimized survey and prompted smartphone use, 57% of respondents used a smartphone as requested. The design features that respondents encounter when they are invited to the survey (invitation mode, incentive) are also key factors in participation. SMS invitations and reminders were both found to be more effective in increasing mobile web response rates than e-mail invitations and reminders (De Bruijne and Wijnant, 2014b; Mavletova and Couper, 2014). Mavletova and Couper (2016b) found that offering differential incentives conditional on the device used for completing the survey can increase response rates among mobile web respondents. They found that offering higher incentives (double the usual amount for panelists) for completing the survey on a mobile phone increased the response rate by 10 percentage points (while including both PC and mobile web respondents) and the proportion of mobile web respondents by 20 percentage points over the control condition offering the usual incentives.

Efforts to *discourage* respondents from using smartphones (e.g., by prompting those using smartphones to switch to a PC), on the other hand, have not been particularly successful (see Chapter 10).

7.3.2 Breakoffs

In addition to lower proportions of sample persons starting a survey when directed to do so on a mobile device, mobile respondents are also less likely to complete the survey. Table 7.2 shows a summary of selected studies reporting differences in breakoff rates between mobile and PC web surveys. Based on a meta-analysis of the studies, the average breakoff rate in mobile web is 13.4%, while in PC web it is 5.5%. The risk of breaking off in mobile web is 2.8 times higher than PC web.

A variety of factors—related to both survey design features and respondent device use—have been found to be associated with higher breakoff rates in mobile web. Based on a meta-analysis of breakoff rates among mobile web respondents, Mavletova and Couper (2015) found that mobile optimization (OR[odds ratios] = 0.71, $p < 0.01$), e-mail invitations (vs. SMS, OR = 0.47, $p < 0.001$), a larger number of reminders (OR for three vs. no reminders = 0.39, $p < 0.01$), using prerecruitment to select those willing to use a mobile device (OR = 0.68, $p < 0.05$), and an opportunity to select the preferred device (vs. being assigned to mobile web in an experimental study, OR = 0.62, $p < 0.05$) decreased the odds of breakoffs. Longer questionnaires (OR for 30-minute vs. 5-minute survey = 1.42, $p = 0.07$) and complex surveys with grids, drop-down questions, images, slider bars, and progress indicators (OR for all five elements vs. no elements = 1.91, $p < 0.001$) increase the odds of breakoffs. Survey length explained the largest proportion of the between-study variance ($R^2 = 0.45$) in breakoffs. They found no significant effect of an opportunity to skip survey questions, incentives, indicating expected survey duration in the invitation, and a scrolling (vs. paging) design. Despite this finding, some experimental studies have shown a modest decrease in breakoff rates among mobile web respondents using scrolling designs (Mavletova and Couper, 2014, 2016a; McGeeney and Marlar, 2013). However, the benefit may not be sufficiently large to offset the benefits of the paging approach for designers.

Breakoff rates differ by technical features of mobile devices. Tablet respondents have lower breakoff rates (Guidry, 2012; Mavletova and Couper, 2014, 2016a; Wells et al., 2013b), while feature phone respondents produce higher breakoff rates than smartphone users (Mavletova and Couper, 2013, 2014, 2016a). Mavletova and Couper (2013, 2014) found that breakoff rates among smartphone respondents differed by operating system: Apple iOS users had the lowest

Table 7.2 Breakoff rates (%) for PC and mobile web users.

	PC web	Mobile web	Optimized for mobile?
Baker-Prewitt (2013) ^{a)}	5	18	No
		13	Yes
Barlas et al. (2015) ^{a)}	1.5	7.5	No
		5.5	Yes
Bosnjak et al. (2013)	6	12	No
Buskirk and Andrus (2014) ^{a)}	28.4	30.9	Yes
Couper and Peterson (2016)	13.4	27.7	No
Guidry (2012)	13	27	No
Horwitz (2014)	11.5	26.8	No
Mavletova (2013) ^{a)}			
Short survey	0.7	16.3	Yes
Long survey	2.3	20.3	Yes
Mavletova and Couper (2013) ^{a)}	2.9	13.6	Yes
Maxl (2014)	38	44	No
McGeeney and Marlar (2013)	0.3	1.3	No
		1.0	Yes
Stapleton (2013)	3.3	15.9	No
		6.3–10.2	Yes
Wells et al. (2013a) ^{a)}	1	4	Yes
Average (and 95% CI) from meta-analysis	5.5 (2.5; 9.6)	13.4 (8.4; 19.5)	

a) Random assignment to device; all others self-selected device.

breakoff rates (2%), while Symbian-based users had the highest breakoff rates (13%). They also found that screen width had a significant effect, with each additional pixel decreasing the odds of breaking off by 0.998 (Mavletova and Couper, 2016b). No effect of screen height was found. There is also some evidence that completing the survey using an Internet cellular service (2G) substantially increases the likelihood of breaking off compared to using Wi-Fi or 3G mobile Internet connection (Mavletova and Couper, 2016a).

There are still relatively few experimental studies exploring the reasons behind the lower response rates and higher breakoff rates in mobile web compared to PC web. Both types of non-response may be higher because of survey design features that differentially affect mobile web users. Other factors—such as the speed, reliability, and type of Internet connection—need to be explored using observational data or paradata. Other possible explanations—such as those relating to context of use and multitasking—are harder to investigate directly, other than through follow-up studies of those who fail to respond or who break off (e.g., Mavletova and Couper, 2016a).

7.3.3 Completion Times

While not directly related to nonresponse, survey completion times may indirectly affect unit nonresponse and willingness to use a mobile device to complete surveys, through greater

perceived response burden. With a few exceptions, a number of studies have found that surveys completed on mobile devices (specifically smartphones) take longer than those completed on PCs (see Couper and Peterson, 2016). This seems to be true whether respondents choose the device or are randomized to device. It also appears to be true both for nonoptimized surveys and those optimized for mobile devices.

Several hypotheses have been advanced for the observed differences in completion times, including connection speed and/or quality, screen size (smaller text taking longer to read or greater need for scrolling), input method (taking longer to select a smaller-sized object on a smartphone), familiarity with the device, and context of use (more multitasking or distractions for smartphone users). Mavletova and Couper (2016b) found that transmission times in a survey in Russia (where many still use feature phones) account for about 28% of the time difference, while Couper and Peterson (2016) found that 25% of the time difference in a student survey in the U.S. could be attributed to transmission time in 2012, while this decreased to 13% in 2013 and 2014. Mavletova and Couper (2016b) found that those who used 2G or 3G connections took significantly longer to complete the survey than those who used Wi-Fi. They also found that each additional 100 pixels in screen size decreased predicted completion time by 0.2 minutes. Mobile optimization should improve response selection, but we still see longer times for optimized surveys. Familiarity and comfort should not be a factor in those studies where users choose the device to use. Couper and Peterson's (2016) analysis of item-level times using client-side paradata suggests that the time required to scroll (especially in grids) may account for much of the difference in within-page response times.

Mavletova and Couper (2013) found that mobile users were more likely to report completing the survey away from home or office (see also Antoun, 2015b; de Bruijne and Wijnant, 2013; Lorch and Mitchell, 2014; Revilla et al., 2014) and more likely to report the presence of others. Antoun (2015b) found that smartphone users were more likely to report multitasking while completing the survey. These differences in self-reported distraction or multitasking may not be sufficiently large to account for the time differences by device. While the effect of device on completion time is well established, the explanations for this difference are not yet fully understood. The effects of the longer completion times on nonresponse, breakoffs, and data quality have also not yet been explored.

7.3.4 Compliance with Special Requests

In order to exploit the capabilities of smartphones, respondents must be willing and able to download and install apps, activate GPS, take photographs, use QR codes, and the like. Noncompliance to these requests may introduce an additional source of nonresponse bias for such surveys. While the use of enhanced smartphone features is widespread in small-scale studies among volunteers, to date there is little research on how well this may work in large-scale surveys of the general population.

Toepoel and Lugtig (2014) asked panelists in the Netherlands for permission to capture their GPS coordinates: 26% of mobile respondents and 24% of PC participants permitted the one-time capture of GPS data. Biler et al. (2013) surveyed people in the Czech Republic about their willingness to participate in a travel survey using a GPS device: only 8% expressed willingness, while 67% said no (the remainder being uncertain). Similarly, Armoogum et al. (2013) asked participants in the 2007–2008 French National Travel Survey about their willingness to accept a GPS device to monitor their travel: 29.8% said yes without condition, 5.1% said yes as long as they could turn it off, and 64.3% said no.

Revilla et al. (2014) asked NetQuest panelists in several countries about their willingness to perform additional survey tasks using their mobile devices. For example, about 36% of Spanish

respondents said “yes, for sure” and a further 27% said “probably” to installing an app on their smartphone, while 15% said “no” (the rest said “maybe”). Similarly, 29% of Spanish respondents said “yes, for sure” to taking pictures using their smartphones, while 19% said “no”; 24% said “yes, for sure” to sharing their GPS locations, while 33% said “no.” The proportion of “no” responses differed across countries and tasks: for installing an app, it ranged from 6% in Mexico to 18% in Argentina; for taking pictures it ranged from 8% in Mexico to 24% in Portugal, and for sharing GPS the proportion of “no” responses ranged from 22% in Mexico to 35% in Argentina.

Crawford et al. (2013) asked student panelists at a U.S. university if they would be willing to allow the capture of their geolocation information: 58% responded positively to the hypothetical question. They conducted a subsequent experiment in which respondents were randomized to receive the consent request for geolocation data as part of the overall survey consent, or as a separate request (a third group was not asked for geolocation data). While 97% of those who opened the survey consented to the survey and GPS capture when asked together, 67% consented when the two were separated. At the time the actual capture occurred, 20% of those in the first group and 33% of those in the second group produced usable geolocation data. Even among trained professionals, the compliance rate may not be optimal. Olson and Wagner (2013) equipped interviewers with GPS-enabled smartphones and asked them to activate an app to track their work-related travel each day. This process yielded GPS files for 59% of the interviewer days.

Ribeiro and Kuegler (2014) reported on a request to take a picture at the end of the survey: 56% of smartphone respondents completed the task, while 36% of feature phone respondents did so. Asking respondents to watch a video had similar negative effects, with between 60 and 70% of respondents failing to make it to the end of the survey where a video was displayed.

In a recent experiment, McGeeney and Weisel (2015) randomly assigned eligible smartphone users to either a mobile web or an app version of experience sampling. While 90% of respondents in the web group agreed to participate, 76% of those in the app group did so, but only 80% of those actually downloaded the app, resulting in compliance from 61% of those invited. App response rates were also significantly lower throughout the seven-day period than browser-based rates.

Much work remains to explore the nonresponse bias implications of using the enhanced features of mobile devices in large-scale surveys. None of the above studies examined differences between those who did and did not comply with these requests. Further research is also needed on the best ways to elicit informed consent for such enhanced features (especially those using passive methods) and enhance compliance rates for such supplemental tasks.

7.4 Measurement Error

Early research on measurement error focused on observational comparisons between mobile and PC web respondents. More recent research has used randomized experiments to compare PC versus mobile users, mobile-optimized versus nonoptimized surveys, and alternative question designs. This research was driven by early concerns that limitations in screen size and processing power, as well as the absence of a mouse and standard keyboard, may increase the measurement error on mobile devices. Most studies show small or no differences in measurement error. Lugtig and Toepoel (2015) suggest that the differences are rather between the respondents who choose either a mobile device or a PC than between the devices.

Though measurement error does not appear to be a primary concern for mobile web surveys, some specific design details should be considered when designing mobile-optimized surveys. Below we review the following issues: grouping of questions, effects of question type, response

and scale effects, and item nonresponse. Table 7.3 summarizes the main findings and offers some suggestions for future research in this area.

7.4.1 Grouping of Questions

7.4.1.1 Question-Order Effects

One hypothesis is that the small screen sizes of mobile devices may change question context effects relative to PC web, especially if multiple items are visible on the same screen. A related hypothesis is that multitasking and other distractions may interfere with the carryover effect from one item to the next, reducing context effects relative to PC web. While Peytchev and Hill (2010) replicated context effects found in other modes in a small sample of users provided with smartphones, we know of no studies that have explored these issues in mobile web versus PC web.

7.4.1.2 Number of Items on a Page

Three experimental studies have examined whether the number of items on a page has an effect on measurement error in mobile web surveys. None found any significant effects. However, a large number of items presented on a screen are associated with higher item nonresponse.

De Bruijne and Wijnant (2014b) compared six items on a page with a single-item-per-page design and found no differences in item means, response-order effects, or item nonresponse rates in a mobile web survey. Mavletova and Couper (2014) found no difference in item nonresponse rates between the paging and scrolling version with eight items on a page. In their subsequent experiment (Mavletova and Couper, 2016a), they compared three conditions with 5, 15, or 30 items presented on a page in a mobile web survey with and without user-controlled skips. They found no differences in response-order effects, nonsubstantive responses, and the length of open-ended answers. However, displaying 30 items on a page produced twice higher item nonresponse rate in the survey with skips and four times higher in the survey without skips compared to displaying 5 items.

7.4.1.3 Grids versus Item-By-Item

Grids pose a particular challenge for mobile web users. Several experiments measured the effect of grids versus alternative designs—either an item-by-item format (scrolling) or a single-item-per-page format (paging). In one of the earliest comparisons, Peytchev and Hill (2010) compared 10 questions with three response options in a grid versus a single-item-per-page format. They found no differences in Cronbach's alpha, item means, or degree of straightlining.

Lattery et al. (2013) compared a grid with an item-by-item format in a mobile-optimized web survey, with respondents self-selecting the device. For the 11-point scale, they found substantially higher agreement in item means between mobile and PC web responses in the item-by-item than in the grid format ($r = 0.95$ vs. $r = 0.51$). No difference was found for the five-point and two-point scales. They also measured if splitting a grid with 10 items into three shorter grids had an effect on data quality, but found no differences in breakoff rates or item means between versions for both mobile and PC web respondents.

McClain and Crawford (2013) compared a single-item-per-page format with a grid format among smartphone respondents. They found higher data quality in the single-item format: fewer breakoffs (32.3% vs. 37.1%, *n.s.*), shorter median completion times, a lower level of straightlining (10.5% vs. 25.5% in the grids, $p = 0.06$), and higher reporting of drinking-related behavior. At the same time, no differences in item missing data rates were found.

Peterson et al. (2013; see also Chapter 10) conducted an experiment with mobile respondents randomly assigned to a standard web survey design with grids or to different versions

Table 7.3 Measurement error in mobile web surveys: main findings and possible directions for further research.

	Main findings	Possible directions for further research
Grouping of questions		
Question-order effects	Similar to other survey modes, the context effect was found in mobile web surveys (Peytchev and Hill, 2010 ^a).	Comparison of context effects between mobile and PC web surveys.
Grouping of items on a page	No effect of the number of items presented on a page: no effect on response-order effects, the rate of nonsubstantive responses, and the length of answers to open-ended questions in mobile web surveys (de Bruijne and Wijnant, 2014b ^a ; Mavletova and Couper, 2016a ^a). A large number of questions presented on a screen (e.g., 30) produce higher item nonresponse rates (Mavletova and Couper, 2016a ^a).	The effect of number of items presented on a page on validity and reliability measures.
Grids versus item-by-item or single-item-per-page format	Some advantages of using an item-by-item or single-item-per-page formats in mobile web surveys. A higher correlation in the item-by-item than in the grid format (Lattery et al., 2013). A lower level of straightlining in the item-by-item or single-item-per-page formats (McClain and Crawford, 2013; Peterson et al., 2013 ^a). Grids in PC web versus single-item per page in PC web versus single-item per page in mobile web: no significant differences between the conditions in item means (de Bruijne and Wijnant, 2013 ^a).	Dealing with grid questions between mobile and PC web respondents: grids for PC and an item-by-item format for mobile versus item-by-item for both survey modes; effects on validity and reliability.
Effects of question type		
Social desirability bias	Few differences between mobile web and PC web. No differences in behavioral or attitude questions (Antoun, 2015b ^a ; Mavletova, 2013 ^a ; Mavletova and Couper, 2013 ^a). A lower level of reporting in cognitively demanding questions (Mavletova and Couper, 2013 ^a).	Repeated measures laboratory experiments with a crossover design.
Open-ended questions	Length of open-ended answers among mobile web respondents: either shorter or similar to PC web respondents. No other differences between mobile and PC web responses (e.g., item nonresponse rates, differences in content) (Bosnjak et al., 2013; Mavletova, 2013 ^a ; Peterson, 2012; Toepoel and Lugtig, 2014). Larger text boxes produce longer open-ended responses between mobile and PC web respondents (Wells et al., 2013a ^a).	The effect of different visual elements to enhance open-ended responses between mobile and PC web respondents (e.g., dynamic text boxes). The effect of having a number of open-ended questions in the survey on length of answers, breakoff rates and item-nonresponse rates among mobile and PC web respondents.

Table 7.3 (continued)

	Main findings	Possible directions for further research
Response and scale effects		
Primacy effects	No difference between mobile and PC web respondents (Mavletova, 2013 ^a ; Wells et al., 2013a ^a).	
Slider bars and drop-down questions	Drop-down questions produce stronger primacy effects than radio buttons or slider bars among mobile web respondents (Peterson et al., 2013 ^a ; Stapleton, 2013). Response distributions in drop-down and slider questions among mobile web respondents differ from PC web responses with radio button questions (Peterson et al., 2013 ^a). Slider questions produce more response errors in mobile web than in PC web (Antoun, 2015b ^a).	Comparison of radio buttons, slider bars, and drop-down questions between mobile and PC web respondents; effect on validity and reliability.
Scale orientation	No differences in scale orientation (vertical vs. horizontal) between mobile and PC web respondents (Chrzan and Saunders, 2012; de Bruijne and Wijnant, 2014b ^a ; Keusch et al., 2014 ^a ; Peytchev and Hill, 2010 ^a).	
Item missing data		
Item missing data	Item nonresponse rates among mobile web respondents either higher or similar to PC web respondents. Higher among feature phone than among smartphone respondents (Mavletova and Couper, 2014 ^a , 2016a ^a , 2016b ^a).	Explaining possible differences in item nonresponse rates between mobile and PC web respondents (e.g., inadvertently missing, multitasking, technical difficulties).

a) Random assignment to device or mobile web survey mode is the only survey mode; all others self-selected device.

of single-item-per-page formats. Mobile respondents were more likely to straightline on one or more of the four grids than in the item-by-item alternatives.

Many survey software systems optimize for mobile web respondents by converting them to item-by-item questions for mobile users, while leaving them as grids for PC users. De Bruijne and Wijnant (2013) measured if optimizing grids in a single-item-per-page format for smartphones means a similar design should be used for PC users. They compared a single-item-per-page format among mobile users to a grid format among mobile users and a grid format among PC web users, and found almost no significant differences in item means.

While these studies suggest some advantage of item-by-item or single-item formats over grids, they also point to the need for more research on how to deal with grid questions, for both mobile web and PC web surveys (see Couper et al., 2013).

7.4.2 Effects of Question Type

7.4.2.1 Socially Undesirable Questions

Mobile phones can be used in a great diversity of social contexts, including public places in the presence of others, which could increase social desirability bias. A competing hypothesis is that a

mobile phone is a personal device typically not shared with others, which can decrease social desirability bias compared to a PC. The research findings suggest no differences between devices in socially desirable responding. However, a lower level of reporting has been found in mobile web surveys in more cognitively demanding questions (e.g., monthly alcohol consumption or household income).

Mavletova (2013) compared mobile web and PC responses to attitude questions toward socially undesirable practices, and found no significant differences. Mavletova and Couper (2013) conducted a two-wave experiment with a crossover design, with respondents answering the same set of questions using each type of device. They found lower levels of reporting of alcohol consumption and household income in the mobile condition, but no differences on a variety of other deviant behaviors and attitude questions. Antoun (2015b) also conducted a two-wave crossover experiment comparing mobile and PC web responses to five sensitive questions, and found no significant differences.

Other studies have reported that self-selected smartphone respondents gave fewer socially desirable answers and provided more negative scores while evaluating some brands and services. Lattery et al. (2013) found lower scores among mobile web respondents on customer satisfaction. Baker-Prewitt (2014) reported consistently lower brand scores among mobile web respondents; however, the difference was not found in a repeated measures laboratory experiment among 61 respondents with a crossover design.

7.4.2.2 Open-Ended Questions

An early concern was that the difficulty of typing on mobile devices would affect responses to open-ended questions. While some researchers report shorter answers among mobile web respondents (Mavletova, 2013; Peterson, 2012; Peterson et al., 2013; Wells et al., 2013a), others find no significant differences (Bosnjak et al., 2013; Buskirk and Andrus, 2014; Toepoel and Lugtig, 2014; Zahariev et al., 2009). Researchers also found no differences in item nonresponse rates, rates of nonsubstantive responses, or substantive differences in the answers between the survey modes (Bosnjak et al., 2013; Mavletova, 2013; Peterson, 2012; Toepoel and Lugtig, 2014). Peytchev and Hill (2010) found that mobile web respondents were less likely to select the “other” response category in a half-open question than in a closed-ended question. However, this finding was not replicated by either Wells et al. (2013a) or de Bruijne and Wijnant (2014b).

Wells et al. (2013a) compared the effect of a small versus a large text box on mobile app and PC web respondents. Consistent with theoretical expectations and earlier web experiments, they found a significant effect of text box size on the length of responses. Both mobile and PC web respondents entered longer open-ended responses in the large text box.

7.4.3 Response and Scale Effects

7.4.3.1 Primacy Effects

Larger primacy effects may be expected on mobile web devices because of later options being less visible due to smaller screen sizes than on PCs. However, researchers have found no difference in primacy effects between mobile and PC web respondents. Mavletova (2013) compared primacy effects in both single-response and multiple-response factual and attitude questions. She found a significant primacy effect in only one of seven questions, with higher primacy effects for PC web respondents. Wells et al. (2013a) compared primacy effects between a PC web and a mobile app survey, and found no significant differences.

7.4.3.2 Slider Bars and Drop-Down Questions

Several studies have explored alternatives to radio buttons in a mobile web survey. Drop-downs were found to produce stronger primacy effects than radio button formats. Response distributions in both drop-down and slider questions were found to differ from those using radio buttons. These findings generally match those from the web survey literature.

Buskirk and Andrus (2014) compared distributions between iPhone and PC web respondents on slider bar questions, and found no significant difference by device. Courtright and colleagues (2013) compared extremeness, acquiescence, and midpoint response styles in a number of scales, using three different versions for mobile users: grids and sliders in a browser-based survey, and sliders in an app-based survey. They found some differences between the formats, but the results were generally mixed.

Peterson et al. (2013) compared different question formats among mobile web respondents: slider bars, drop-downs, and numeric entry. The drop-down questions were implemented using the native features of each device, meaning that some (e.g., Apple's iOS) presented these using a picker⁴ tool. They found that the response distributions in the mobile slider condition differed the most from the PC web condition, with drop-down questions also differing from the PC web responses (see also Chapter 10). Drop-down questions produced higher primacy effects than the other mobile web conditions.

Stapleton (2013) also found higher primacy effects in drop-down questions than in radio button scales among mobile web respondents. However, Keusch et al. (2014) compared drop-down with radio button questions between iPhone and PC web respondents and found no significant differences. Antoun (2015b) found that mobile web users provided less accurate responses than PC web users when moving a slider bar to record their age (a verifiable measure). He also compared response errors among mobile web respondents using a date picker and PC web respondents using a drop-down menu to record the year of birth, and found more errors and more missing data among mobile web respondents using the date picker.

While more research is needed on this topic, it appears that some of the scale formats designed specifically for mobile use (e.g., sliders, drop boxes, or spinners) do not perform as well as traditional radio button scales when compared to PC web responses.

7.4.3.3 Scale Orientation

Several experiments compared scale orientation (vertical vs. horizontal) among mobile web respondents. The general finding is that neither scale orientation consistently produces better data quality.

Peytchev and Hill (2010) compared horizontal and vertical orientations of a five-point scale. They found no difference in primacy effects, level of differentiation, item means, or Cronbach's alpha coefficients between the conditions. Chrzan and Saunders (2012) compared test-retest reliability in horizontally and vertically aligned scales between mobile and PC web respondents. They found slightly higher test-retest correlations in the horizontal than in the vertical scales (0.73 and 0.65, respectively) for mobile and PC respondents combined. De Bruijne and Wijnant (2014b) compared the effect of scale orientation among mobile web respondents and found no difference in item means between the conditions. However, they found higher item nonresponse rates in the horizontal than in the vertical scale orientation. Keusch et al. (2014) compared scale direction (high to low and low to high) and scale alignment (vertical vs. horizontal) between iPhone and PC web respondents, and found no significant difference in item means.

⁴ A picker is a rotating wheel that the user spins to make a selection. See http://www.w3schools.com/jquerymobile/jquerymobile_form_select.asp.

7.4.4 Item Missing Data

Given the smaller display sizes and the greater need for scrolling on mobile devices, there is a concern that more items may be inadvertently missed. However, there is mixed evidence that item nonresponse rates are higher among mobile web than PC web respondents.

Buskirk and Andrus (2012) and Wells et al. (2013a) found similar item missing data rates in mobile-optimized and PC surveys. However, Mavletova and Couper (2014, 2016a, 2016b) found higher item nonresponse rates for mobile than PC web respondents. Item nonresponse rates among smartphone respondents were found to be three times higher, and among feature phone respondents nine times higher, than among PC web respondents.

While some researchers find no significant difference between tablet and smartphone respondents (Guidry, 2012; Wells et al., 2013b), others find lower item nonresponse rates among tablet than smartphone users (Mavletova and Couper, 2014).

7.5 Links Between Different Error Sources

Some design options in mobile web surveys affect multiple error sources simultaneously, often leading to complicated tradeoffs between such errors. This is most apparent with the questionnaire design features that affect both measurement and nonresponse (through breakoffs). As noted earlier, a paging design may improve measurement by reducing item nonresponse as it does in PC web. On the other hand, it has consequences for nonresponse because it may produce more breakoffs than a scrolling design. Similarly, grids can shorten completion times but they tend to produce more breakoffs than an item-by-item format. In addition, interactive input tools (sliders, spinners, drop-down menus) are often utilized in an attempt to improve measurement, but their use may backfire and increase the likelihood of breakoffs or item nonresponse.

Error tradeoffs are also apparent when choosing between browser-based versus app-based approaches. The latter can extend measurement by taking full advantage of a smartphone's capabilities. But this approach has coverage implications if those who do not use mobile apps are excluded and nonresponse implications if some people are unwilling to download such apps.

The choice of e-mail versus SMS as a mode of invitation for mobile web surveys also presents an error tradeoff. E-mail invitations may improve coverage because they are more likely to be received than SMS invitations. On the other hand, e-mail invitations tend to produce lower response rates than SMS invitations for mobile web surveys.

The amount of time and number of sessions used to complete surveys can also affect multiple sources of errors. There is evidence that mobile surveys are significantly longer and are completed over more sessions than PC surveys. This can affect both nonresponse and measurement error, often in the same way. Longer completion times in mobile may be a deterrent not only to participation (affecting nonresponse error) but also to careful, conscientious responding (affecting measurement error). Meanwhile, completing surveys over multiple sessions (frequent starts and stops) may affect the likelihood of breakoffs and also have unintended consequences for measurement.

Given these links between different error sources, it is still an open question whether mobile administration affects one source of error more than the others. In an experimental study using the LISS panel, Antoun et al. (2014) estimated mobile versus PC web mode effects across seven questions (the PC web survey was used as a benchmark). When they assumed that those without their own smartphone could not participate because they were not covered, the main error source was noncoverage (on average 2.4%) rather than nonresponse (1.3%) or measurement bias

(1.5%). However, in reality, the panel took the extra step of providing phones to those who did not have them, and this strategy had an impact on multiple error sources. It eliminated coverage errors by definition, and reduced nonresponse bias but had a small unintended effect of inflating the measurement bias (to 1.9%). While this is a modest increase, it suggests that those who received smartphones may have had less familiarity and comfort with their devices. This result suggests the need for more research not only about the magnitude of different errors in mobile web but also about the effects of interventions to reduce such errors.

7.6 The Future of Mobile web Surveys

Considerable research has focused on mobile web surveys in recent years. Initially, this focused on differences observed between those who chose to use mobile devices to complete web surveys and those who used PCs or tablets to do so. Subsequent attention turned to experiments designed to disentangle the selection effects from the effects of the device used on data quality. More recently, research has focused on ways to encourage greater use of mobile devices, and on the effects of mobile optimization on both nonresponse and measurement error. We see a maturity in the research on mobile web with research now turning to increasingly sophisticated experiments designed to test theoretically derived hypotheses regarding the early differences observed. Despite the progress made in understanding the mobile web phenomenon, there remain a number of gaps in our knowledge about how to take advantage of this new medium, or how to minimize the negative impacts it may have on survey taking.

One interesting paradox is that, while smartphone penetration is increasing around the world, and that in many countries large majorities of the population have access to mobile devices, relatively small proportions of sample persons choose to complete web surveys on their mobile devices. One possible explanation is that people who have multiple devices choose different devices for different activities. For instances, users may prefer mobile devices for consuming information (e.g., reading e-mail and checking Facebook updates), while using other devices (PCs or tablets) more for production of information or content (e.g., sending e-mails and posting updates to social media), with survey completion falling more in the latter category than in the former. Efforts to increase the proportion of mobile web completes—whether through differential incentives, SMS invitations, shorter surveys, mobile optimization, or the like—show promise, but mobile survey participation rates are still well below smartphone penetration rates. This remains a challenge, both for regular web surveys as the mobile-only or mobile-mostly population continues to grow, but also for surveys designed exclusively to take advantage of the advanced features offered by mobile devices.

A consistent-related finding is that mobile web surveys have lower response rates, higher breakoff rates, and longer completion times than PC web surveys. Again, research focused on optimizing design to accommodate mobile users has made some progress but has not eliminated this gap. Understanding the reasons behind these differences is a critical step in figuring out how to deal with them. Early work on modular survey design for mobile web (e.g., Johnson et al., 2012; Kelly and Stevens, 2013; Siluk et al., 2013) shows some promise, but figuring out the trade-offs for both mobile users and PC users remains.

A clear positive set of findings in the literature is that there are generally few measurement error differences between those who complete a web survey on a mobile device or on a PC. This is especially true if surveys are optimized for use on mobile devices. But many of the design features that have proven to be problematic in web surveys in general also appear to be a challenge for mobile users. These include grid questions, horizontal scrolling, and complex question

types (drop boxes or spinner boxes, sliders, etc.). While there is as yet no evidence of *improvements* in data quality using the advanced features available on mobile devices, it is at least reassuring that well-designed web surveys optimized for mobile devices produce data of comparable quality to that from PCs. However, more work is needed to understand *why* some of the differences observed occur, and *how* the mobile survey experience differs from that on PCs. Disentangling technology effects (e.g., transmission speed and reliability of the connection) from respondent behavior (e.g., multitasking) and characteristics of the device (e.g., display size and input method) is important in advancing our understanding of mobile surveys, and allows us to generalize beyond the current generation of devices. For instance, understanding the difference between smartphone and tablet use will help us anticipate the likely effect of new classes of devices (like phablets) on survey error.

An additional challenge is that of exploiting the advanced features of mobile devices to extend measurement. A lot of promising developments are being tested among volunteers, but extending this to large-scale surveys of the general population remains a challenge. Evidence on respondents' willingness to capture GPS, download apps, take photos, and so on shows that we have a long way to go to realize the full potential of mobile devices. Ethical and legal issues arise with the passive collection of GPS data, while active measurement raises consent and compliance issues.

In summary, survey researchers have made great strides in adapting to the technological challenges presented by the increased use of mobile devices. Understanding how to mitigate the negative effects of mobile device use on survey completion and how to make best use of the measurement opportunities offered by mobile technologies remains a key part of the research agenda.

References

- AAPOR (2014). *Mobile technologies for conducting, augmenting and potentially replacing surveys: Report of the AAPOR task force on emerging technologies in public opinion research*. Deerfield: American Association for Public Opinion Research.
- Antoun, C. (2014). A look at the causes of nonresponse in a mobile-web survey. Paper presented at the WebDataNet conference on mobile research, Larnaca, Cyprus, March 31.
- Antoun, C. (2015a). Who are the internet users, mobile internet users, and mobile-mostly internet users? Demographic differences across internet-use subgroups in the U.S. In D. Toninelli, R. Pinter, and P. de Pedraza (eds) *Mobile research methods: Opportunities and challenges of mobile research methodologies*. London: Ubiquity Press.
- Antoun, C. (2015b). Effects of mobile versus PC web on survey response quality: A crossover experiment in a probability web panel. Paper presented at the AAPOR annual conference, Hollywood, FL, May 14–17.
- Antoun, C. and Couper, M.P. (2013). Mobile-mostly internet users and selection bias in traditional web surveys. Paper presented at the annual meeting of the Midwest Association for Public Opinion Research, Chicago, IL, November 22–23.
- Antoun, C., Couper, M.P., and Conrad, F.G. (2014). Comparing errors from non-coverage to other errors in a mobile web survey. Paper presented at the Internet Survey Methodology workshop, Bozen-Bolzano, Italy, December 1–3.
- Armoogum, J., Roux, S., and Pham, T.H.T. (2013). Total nonresponse of a GPS-based travel survey. Paper presented at the conference on New Techniques and Technologies for Statistics, Brussels, Belgium, March 5–7.

- Baker-Prewitt, J. (2013). Mobile research risk: What happens to data quality when respondents use a mobile device for a survey designed for a PC. Paper presented at the CASRO Online Research conference, San Francisco, CA, March 6–8.
- Baker-Prewitt, J. (2014). Do smartphones really produce lower scores? Understanding device effects on survey ratings. Paper presented at the CASRO Digital Research conference, San Antonio, TX, March 11–12.
- Barlas, F.M., Thomas, R.K., and Graham, P. (2015). Mobility enabled: Effects of mobile devices on survey response and substantive measures. Paper presented at the CASRO Digital Research conference, Nashville, TN, February 11–12.
- Biler, S., Šenk, P., and Winklerová, L. (2013). Willingness of individuals to participate in a travel behavior survey using GPS devices. Paper presented at the conference on new techniques and technologies for statistics, Brussels, Belgium, March 5–7.
- Bosnjak, M., Metzger, G., and Gräf, L. (2010). Understanding the willingness to participate in mobile surveys: Exploring the role of utilitarian, affective, hedonic, social, self-expressive, and trust-related factors. *Social Science Computer Review*, 28, 3, 350–370.
- Bosnjak, M., Poggio, T., Becker, K.R., Funke, F., Wachenfeld, A., and Fischer, B. (2013). Online survey participation via mobile devices. Paper presented at the AAPOR annual conference, Boston, MA, May 16–19.
- Buskirk, T.D. and Andrus, C. (2012). Smart surveys for smart phones: Exploring various approaches for conducting online mobile surveys via smartphones. *Survey Practice*, 5, 1.
- Buskirk, T.D. and Andrus, C. (2014). Making mobile browser surveys smarter: Results from a randomized experiment comparing online surveys completed via computer or smartphone. *Field Methods*, 26, 4, 322–342.
- Callegaro, M. and DiSogra, C. (2008). Computing response metrics for online panels. *Public Opinion Quarterly*, 72, 5, 1008–1032.
- Carley-Baxter, L.R., Peytchev, A., and Black, M.C. (2010). Comparison of cell phone and landline surveys: A design perspective. *Field Methods*, 22, 3–15.
- Chrzan, K. and Saunders, T. (2012). Improving the mobile survey experience: A closer look at scale orientation, grids and modality effects. Maritz research whitepaper. <http://blog.maritz.de/wp-content/uploads/2014/02/%E2%80%9EScale-Orientation%E2%80%99C1.pdf> (accessed May 6, 2016).
- Couper, M.P., and Peterson, G. (2016). “Why Do Web Surveys Take Longer on Smartphones?” *Social Science Computer Review*, available online, DOI: 10.1177/0894439316629932.
- Couper, M.P., Tourangeau, R., Conrad, F., and Zhang, C. (2013). The design of grids in web surveys. *Social Science Computer Review*, 31, 3, 322–345.
- Courtright, M., Pashupati, K., Pettit, A., and Knowles, R. (2013). Multimode, global scale usage: Understanding respondent scale usage across borders and devices. *Proceedings of ESOMAR Congress*, September 22–25, Istanbul.
- Crawford, S.D., McClain, C., Young, R.H., and Nelson, T.F. (2013). Understanding mobility: Consent and capture of geolocation data in web surveys. Paper presented at the AAPOR annual conference, Boston, MA, May 16–19.
- Cui, Y. and Roto, V. (2008). How people use the web on mobile devices. *Proceedings of the 17th International Conference on World Wide Web*, April 21–25, ACM, New York, 905–914.
- De Bruijne, M. and Wijnant, A. (2013). Comparing survey results obtained via mobile devices and computers: An experiment with a mobile web survey on a heterogeneous group of mobile devices versus a computer-assisted web survey. *Social Science Computer Review*, 31, 4, 482–504.
- De Bruijne, M. and Wijnant, A. (2014a). Mobile response in web panels. *Social Science Computer Review*, 32, 6, 728–742.

- De Bruijne, M. and Wijnant, A. (2014b). Improving response rates and questionnaire design for mobile web surveys. *Public Opinion Quarterly*, 78, 4, 951–962.
- Duggan, M. and Smith, A. (2013). *Cell internet use 2013*. Washington, DC: Pew Internet and American Life Project.
- Fuchs, M. and Busse, B. (2009). The coverage bias of mobile web surveys across European countries. *International Journal of Internet Science*, 4, 1, 21–33.
- Guidry, K.R. (2012). Response quality and demographic characteristics of respondents using a mobile device on a web-based survey. Paper presented at the AAPOR annual conference, Orlando, FL, May 17–20.
- Horwitz, R. (2014). Usability of the ACS internet instrument on mobile devices. Paper presented at Statistics Canada's international methodological symposium, Hull, Quebec, Canada, October 29–31.
- Johnson, A., Kelly, F., and Stevens, S. (2012). Modular survey design for mobile devices. Paper presented at CASRO Online Research conference, Las Vegas, NV, March 1–2.
- Kelly, F. and Stevens, S. (2013). Bite-sized chunks'—Mobile and CAWI parallel proposal. Paper presented at the CASRO Online Research conference, San Francisco, CA, March 6–8.
- Kelly, F., Johnson, A., and Stevens, S. (2013). Modular survey design: Bite sized chunks 2. Paper presented at the CASRO Online Research conference, San Francisco, CA, March 6–8.
- Keusch, F., Yan, T., Han, S., and He, L. (2014). Impact of response scale direction on survey responses in web and mobile web surveys. Paper presented at the General Online Research conference, Cologne, Germany, March 5–7.
- Lattery, K., Park Bartolone, G., and Saunders, T. (2013). Optimizing surveys for smartphones: maximizing response rates while minimizing bias. Paper presented at the CASRO Online Research conference, San Francisco, CA, March 6–8.
- Lipsman, A. and Aquino, C. (2013). Mobile future in focus. comScore Report. <http://www.comscore.com/Insights/Presentations-and-Whitepapers/2013/2013-Mobile-Future-in-Focus> (accessed May 6, 2016).
- Lorch, J. and Mitchell, N. (2014). Why you need to make your surveys mobile friendly now. Shelton: Survey Sampling International, webinar slides, downloaded May 24, 2014.
- Lutig, P. and Toepoel, V. (2015). The use of PCs, smartphones, and tablets in a probability-based panel survey: Effects on survey measurement error. *Social Science Computer Review*. doi: 10.1177/0894439315574248.
- Mavletova, A. (2013). Data quality in PC and mobile web surveys. *Social Science Computer Review*, 31, 6, 725–743.
- Mavletova, A. and Couper, M.P. (2013). Sensitive topics in PC web and mobile web surveys: Is there a difference? *Survey Research Methods*, 7, 3, 191–205.
- Mavletova, A. and Couper, M.P. (2014). Mobile web survey design: Scrolling versus paging, SMS versus e-mail invitations. *Journal of Survey Statistics and Methodology*, 2, 4, 498–518.
- Mavletova, A. and Couper, M.P. (2015). A meta-analysis of breakoff rates in mobile web surveys. In D. Toninelli, R. Pinter, and P. de Pedraza (eds) *Mobile research methods: Opportunities and challenges of mobile research methodologies*. London: Ubiquity Press.
- Mavletova, A. and Couper, M.P. (2016a). Grouping of items in mobile web questionnaires. *Field Methods*, 28, 2, 170–193.
- Mavletova, A. and Couper, M.P. (2016b). Device use in web surveys: The effect of differential incentives. *International Journal of Market Research*, 58, 4, 523–544.
- Maxl, E. (2014). Students first choice: The influence of mobile mode on results. Paper presented at the General Online Research conference, Cologne, Germany, March 5–7.
- McClain, C. and Crawford, S.D. (2013). Grid formats, data quality, and mobile device use: A questionnaire design approach. Paper presented at the AAPOR annual conference, Boston, MA, May 16–19.

- McGeeney, K. and Marlar, J. (2013). Mobile browser web surveys: Testing response rates, data quality, and best practices. Paper presented at the AAPOR annual conference, Boston, MA, May 16–19.
- McGeeney, K. and Weisel, R. (2015). App vs. web for surveys of smartphone users experimenting with mobile apps for signal-contingent experience sampling method surveys, Pew Research Center report, Washington, DC. <http://www.pewresearch.org/2015/04/01/app-vs-web-for-surveys-of-smartphone-users/> (accessed May 6, 2016).
- Metzler, A. and Fuchs, M. (2014). Coverage error in mobile web surveys across European countries. Paper presented at the Internet Survey Methodology workshop, Bozen-Bolzano, Italy, December 1–3.
- Millar, M.M. and Dillman, D.A. (2012). Encouraging survey response via smartphones: Effects on respondents' use of mobile devices and survey response rates. *Survey Practice*, 5, 2.
- Olson, K. and Wagner, J. (2013). A field experiment using GPS devices to measure interviewer travel behavior. Paper presented at the AAPOR annual conference, Boston, MA, May 16–19.
- Pearce, K.T. and Rice, R.E. (2013). Digital divides from access to activities: Comparing mobile and personal computer internet users. *Journal of Communication*, 63, 721–744.
- Peterson, G. (2012). Unintended mobile respondents. Paper presented at the CASRO Technology conference, New York, May 31.
- Peterson, G., Mechling, J., LaFrance, J., Swinehart, J., and Ham, G. (2013). Solving the unintentional mobile challenge. Paper presented at the CASRO Online Research conference, San Francisco, CA, March 6–8.
- Peytchev, A. and Hill, C.A. (2010). Experiments in mobile web survey design. *Social Science Computer Review*, 28, 3, 319–335.
- Poggio, T., Bosnjak, M., and Weyandt, K. (2015). Survey participation via mobile devices in a probability-based online-panel: Prevalence, determinants, and implications for nonresponse. *Survey Practice*, 8, 2.
- Revilla, M., Toninelli, D., Ochoa, C., and Loewe, G. (2014). Do online access panels really need to allow and adapt surveys to mobile devices? RECSM working paper no. 41, Universitat Pompeu Fabra, Barcelona. http://www.upf.edu/survey/_pdf/RECSM_wp041.pdf (accessed May 6, 2016).
- Ribeiro, E. and Kuegler, S. (2014). Influential factors on survey outcomes: Length of survey, device selection and external elements. Lightspeed GMI whitepaper. <http://www.lightspeedgmi.com/resource-center/newsletter-2014/white-paper-completion-rates-on-mobile-devices/> (accessed May 6, 2016).
- Robinson, O. (2014). Finding simplicity in a multi-device world. GfK. <http://blog.gfk.com/2014/03/finding-simplicity-in-a-multi-device-world> (accessed May 6, 2016).
- Sarraf, S., Brooks, J., Cole, J., and Wang, X. (2015). What is the impact of smartphone optimization on long surveys? Paper presented at the AAPOR annual conference, Hollywood, FL, May 14–17.
- Saunders, T. and Kessler, A. (2015). Read me! Click me! Innovations in email invitation design for today's digital world. Paper presented at the CASRO Digital Research conference, Nashville, TN, February 11–12.
- Siluk, L., Johnson, E.P., and Tarraf, S. (2013). Cyborgs vs. monsters: Assembling modular surveys to create complete datasets. Paper presented at the CASRO Online Research conference, San Francisco, CA, March 7–8.
- Smith, A. (2015). U.S. smartphone use in 2015. Pew Internet and American Life Project. <http://www.pewinternet.org/2015/04/01/us-smartphone-use-in-2015/> (accessed July 11, 2016).
- Stapleton, C.E. (2013). The smartphone way to collect survey data. *Survey Practice*, 6, 2.
- Statistics Netherlands (2013). Smartphones used most for mobile internet access. *Web Magazine*. <http://www.cbs.nl/> (accessed May 6, 2016).
- Steenson, M. and Donner, J. (2009). Beyond the personal and private: Modes of mobile phone sharing in urban India. In R. Ling and S.W. Campbell (eds) *The reconstruction of space and time: Mobile communication practices*, 231–250. New Brunswick: Transaction Publishers.

- Toepoel, V. and Lugtig, P. (2014). What happens if you offer a mobile option to your web panel? Evidence from a probability-based panel of internet users. *Social Science Computer Review*, 32, 4, 544–560.
- Wells, T., Bailey, J., and Link, M.W. (2013a). Comparison of smartphone and online computer survey administration. *Social Science Computer Review*, 32, 2, 238–255.
- Wells, T., Bailey, J., and Link, M.W. (2013b). Filling the void: Gaining a better understanding of tablet-based surveys. *Survey Practice*, 6, 1.
- Zahariev, M., Ferneyhough, C., and Ryan, C. (2009). Best practices in mobile research. Paper presented at the ESOMAR online research, Chicago, IL, October 26–28.